

ASSIGNMENT - 3

Q 1: Binary Encoding means presenting decisions using binary symbols aka 0-1. In Portfolio Optimization we use:

- To represent whether invest or not
- The level of allocation
- A discretized value of weight
- Or Some firms representation

② GRAY CODE: Binary numerals only where only one bit changes between consecutive numbers.

$$\text{GRAY BIT} \Rightarrow \text{Prev Binary Bit} \oplus \text{Current Binary Bit}$$

XOR

③ (a) On going from 3 → 4:

Bin: 011 → 100 ⇒ 3 bits changed

Gray: 010 → (110) ⇒ 1 bit changed.

(b) Hamming Distance: Num of places they differ.

Bin: Large/Variable [Sometimes all change]

Gray: ~~Always~~ Always 1 [as we defined it].

Binary is faster to Decode.

Gray has smoother Search.

④

$$S_1 \Rightarrow 000101 \Rightarrow 000111 \Rightarrow 5$$

$$S_2 \Rightarrow 001111 \Rightarrow 001000 \Rightarrow 15$$

$$S_3 \Rightarrow 010010 \Rightarrow 011011 \Rightarrow 18$$

for S_1 :

$$g_1 = 0, g_2 = 0 \oplus 0 = 0$$

$$g_3 = 0 \oplus 0 = 0$$

$$g_4 = 0 \oplus 1 = 1$$

$$g_5 = 1 \oplus 0 = 1$$

$$g_6 = 1 \oplus 0 = 1$$

$$\Rightarrow 000111$$

⑤

$$S_4 \Rightarrow 111010 \Rightarrow 101100 \Rightarrow 44$$

$$S_5 \Rightarrow 101101 \Rightarrow 110110 \Rightarrow 54$$

$$S_6 \Rightarrow 010111 \Rightarrow 011010 \Rightarrow 26$$

Bin bit $\Rightarrow$ Prev Bin $\oplus$ Gray Bit

$$S_4 \Rightarrow 0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 0 = 0$$

$$(101100)$$

$$S_5 \Rightarrow 0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$

$$(110110)$$

$$\begin{aligned}
 S_6 \Rightarrow & \begin{aligned} 0 \oplus 0 &= 0 \\ 0 \oplus 1 &= 1 \\ 1 \oplus 0 &= 1 \\ 1 \oplus 1 &= 0 \\ 0 \oplus 1 &= 1 \\ 1 \oplus 1 &= 0 \end{aligned} \Rightarrow 011010.
 \end{aligned}$$

$$\begin{aligned}
 S_2 = & \begin{aligned} 0 \oplus 0 &= 0 \\ 0 \oplus 0 &= 0 \\ 0 \oplus 1 &= 1 \\ 1 \oplus 1 &= 0 \\ 0 \oplus 1 &= 0 \\ 1 \oplus 1 &= 0 \end{aligned} \\
 \Rightarrow & 001000
 \end{aligned}$$

$$\begin{aligned}
 S_3 = & \begin{aligned} 0 \oplus 0 &= 0 \\ 0 \oplus 1 &= 1 \\ 1 \oplus 0 &= 1 \\ 0 \oplus 0 &= 0 \\ 0 \oplus 1 &= 0 \\ 1 \oplus 0 &= 1 \end{aligned} \\
 & 011011
 \end{aligned}$$

⑥ $\Rightarrow$

	%	$f(x)$	Prob	
S_1	10	100	0.0103	0.0103
S_2	25	625	0.0641	0.0744
S_3	30	900	0.0928	0.1667
S_4	45	2025	0.2077	0.3744
S_5	50	2500	0.2564	0.6308
S_6	60	3600	0.3692	1.0000

9750

$$\downarrow \\
 \left[\frac{F(x)}{9750} \right]$$

$$\hookrightarrow P_{i+1} = P_i + P_{i+1}$$

$$\begin{aligned}
 \Rightarrow & \begin{array}{ccccccc} 0.12 & 0.35 & 0.58 & 0.77 & 0.89 & 0.95 \\ S_3 & S_4 & S_5 & S_6 & S_6 & S_6 \end{array}
 \end{aligned}$$

⑥:

$$T_1 [S_1, S_4] = S_4$$

$$T_2 [S_2, S_5] = S_5$$

$$T_3 [S_3, S_6] = S_6$$

$$T_4 [S_1, S_3] = S_3$$

$$T_5 [S_2, S_4] = S_4$$

$$T_6 [S_5, S_6] = S_6$$

⑦ CROSS OVER:

①

$$P_1 = \underline{101} \underline{101} \Rightarrow \underline{101010}$$

$$P_2 = \underline{011} \underline{010} \Rightarrow \underline{011101}$$

②

$$P_1 = 1 \underline{1001} 1 \Rightarrow 1 \underline{0110} 1$$

$$P_2 = 0 \underline{0110} 1 \Rightarrow 0 \underline{1001} 1$$

③

$$P_1 = 101010$$

$$P_2 = 010101$$

$$M = 110010$$

$$m_1 = 1 \Rightarrow 1$$

$$m_2 = 0 \Rightarrow 0$$

$$m_3 = 0 \Rightarrow 0$$

$$m_4 = 0 \Rightarrow 1$$

$$m_5 = 1 \Rightarrow 1$$

$$m_6 = 0 \Rightarrow 0$$

$$\Rightarrow 100111$$

$$\textcircled{d} \quad \begin{array}{l} O_1 = 101010 \Rightarrow P_1 = 101101 \\ O_2 = 011101 \quad P_2 = 011010 \end{array}$$

$$\textcircled{e} \quad \begin{array}{l} P_1 = 111000 \Rightarrow 111111 = O_1 \\ P_2 = 000111 \quad 000000 = O_2 \end{array}$$

$$\hookrightarrow \text{Mutate Bit-2} \Rightarrow 101111$$

$\textcircled{8}:-$

(a) Generate a random number from Uniform distribution U
 $u \sim U(0,1)$ and apply the Inverse CDF
 as $n = F^{-1}(u)$

And Now uniform numbers are converted to
 realistic distribution corresponding to $CDF = F(n)$.

(b)

u	$n = \sqrt{u}$
0.1	$\rightarrow 0.316$
0.25	$\rightarrow 0.500$
0.5	$\rightarrow 0.707$
0.75	$\rightarrow 0.866$
0.9	$\rightarrow 0.949$

③ $\eta = 0.4, \eta_{\min} = 0, \eta_{\max} = 1$

$$Am = 2.$$

$$\delta = (2\eta)^{1/3} - 1.$$

$$\& X_{\text{New}} = X_{\text{old}} + \delta.$$

• $0.2 \Rightarrow \delta = (2 \times 0.2)^{1/3} - 1 = -0.263$

$\hookrightarrow \cancel{0.263} \cdot 0.137.$

• $0.5 \Rightarrow \delta = (2 \times 0.5)^{1/3} - 1 = 0$

$\hookrightarrow 0.5$

• $0.8 \Rightarrow \delta = (2 \times 0.8)^{1/3} - 1 \Rightarrow 0.170$

$\hookrightarrow 0.570.$

$$0.2 \Rightarrow 0.137$$

$$0.5 \Rightarrow 0.5$$

$$0.8 \Rightarrow 0.570$$